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Estimating the benefits of recreation-oriented management in state-owned commercial forests in Finland: A choice experiment

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ABSTRACT

As forestlands provide a variety of environmental services, the management of forest resources is a matter of public concern. In the present case of state-owned commercial forests in Finland, legislation requires specific management practices to enhance recreational benefits free of charge to the public. This choice experiment considers Finnish people's valuation of the recreation-oriented management of state-owned commercial forests to evaluate whether the recreational benefits produced justify the related loss of profits from timber sales. We focus on three management attributes: scenic buffer zones along lakes and rivers, habitats for game birds, and the quality of scenery as reflected by the frequency of clear-cut areas along hiking trails. Marginal willingness-to-pay (WTP) effects for the attributes are estimated with random parameters logit models specified in the WTP space, while preference-space models are used to estimate in physical terms the attribute levels that maximize the benefits to the public. Despite regional differences in preferences, people in all parts of Finland valued the current recreation-oriented management of state-owned commercial forests considerably. Nationwide, the aggregate benefits of recreation-enhancing management clearly exceeded the estimated opportunity costs. The most preferred

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levels of management attributes were slightly above the current levels, suggesting an increase in the provision of recreational services when not considering the associated costs.

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Introduction

Healthy forest ecosystems provide a wide variety of marketed as well as non-marketed goods and services, including timber and other material products, recreational opportunities, wildlife habitat, esthetically pleasing landscapes, and watershed protection. Accordingly, the management of forest resources is a matter of increasing public concern. A consequence is that forest managers and forest policy makers can greatly benefit from research results concerning the general public's or recreationists' preferences for management-related forest characteristics. During the last two decades, attribute-based stated choice methods such as choice experiments (CE) have been found to be particularly useful for resource management in considering the forest management regime as a set of attributes that have management or policy relevance (e.g., [Bennett and Adamowicz, 2001](#); [Holmes and Adamowicz 2003](#); [Holmes and Boyle, 2003](#)). By including a monetary attribute and varying several attributes in the choice tasks, a multidimensional valuation surface can be estimated and value trade-offs among the management attributes measured.

A number of CE studies of forest recreation and environmental benefits of forestlands have been conducted so far. Apart from a monetary attribute (cost of access, tax increase), these have focused on management attributes such as size and shape of harvesting gaps or evidence of forestry activity, especially clear-cuts ([Hanley et al., 1998](#); [Boxall and Macnab, 2000](#); [Holmes and Boyle, 2003](#); [Tyrväinen et al., 2014](#)), forest scenery related to management intensity ([Horne et al., 2005](#); [Christie et al., 2007](#)) or slash disposal ([Holmes and Boyle, 2003](#)), recreational access within the forest ([Adamowicz et al., 1998](#); [Boxall and Macnab, 2000](#)) or access to private forests ([Campbell et al., 2014](#)), as well as tree species and trees left after harvest ([Hanley et al., 1998](#); [Holmes and Boyle, 2003](#); [Campbell et al., 2014](#)). Management attributes considered also include opportunity to see, and populations of, wildlife species ([Adamowicz et al., 1997, 1998](#); [Boxall and Macnab, 2000](#)), hunting and fishing ([Schaberg et al., 1999](#)), area available for timber production or left aside ([Adamowicz et al., 1998](#); [Holmes and Boyle, 2003](#)), measures of biological diversity ([Garrod and Willis, 1997](#); [Adamowicz et al., 1998](#); [Horne et al., 2005](#); [Meyerhoff et al., 2009](#); [Thiene et al., 2012](#); [Tyrväinen et al., 2014](#)), recreational vs. ecological management ([Juutinen et al., 2011](#)), as well as riparian buffer zones and watershed protection ([Holmes and Boyle, 2003](#); [Schaberg et al., 1999](#); [Haefele and Loomis, 2001](#)).²

The present choice experiment study explores Finnish people's preferences for the recreation-oriented management of state-owned commercial forests (SOCFs) in Finland. Overall, state-owned forests cover about one third of Finland's forestry land and amount to 9.2 million hectares, a half of which is commercially managed forest ([Fig. 1](#)). The other half consists of wilderness areas, national parks, and other protected areas. These lands are managed by Metsähallitus, a state-run enterprise that remits the profits from forestry as well as other business operations to the government. In addition, Metsähallitus has public administration duties, such as the management of conservation areas.

Legislation poses specific requirements for the managers of SOCFs to provide environmental services along with timber ([Act on Metsähallitus, 2004](#)). Similar statutes exist, for example, in the USA ([National Forest Management Act, 1976](#)). Finland's SOCFs are estimated to host over 10 million close-to-home recreational visits annually ([Sievänen and Neuvonen, 2011](#)). To enhance recreation,

² [Lindhjem \(2007\)](#) provides a review of stated preference studies in Fennoscandia. For revealed preference data, [Thiene and Scarpa \(2009\)](#) consider site selection for alpine outdoor recreation, with attributes such as conifer vs. broadleaved forests. [Zandersen and Tol \(2009\)](#) provide a meta-analysis of European travel cost studies.

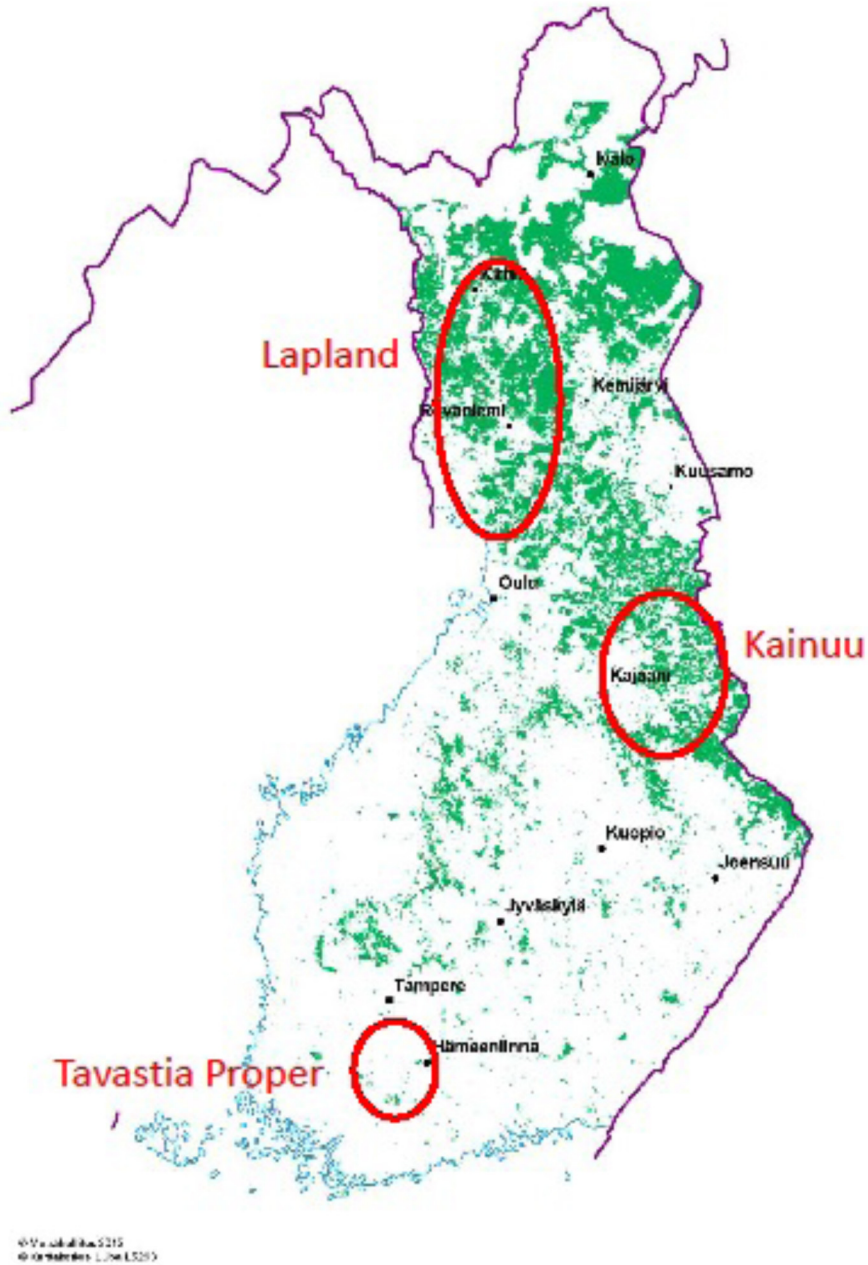


Fig. 1. The focus of this study: the state-owned commercial forests (SOCFs) in Finland and three sampling regions.

Metsähallitus applies specific management practices that are not typical of commercial forestry. For example, buffer zones are left along lakes, rivers, and hiking trails to preserve wooded scenery. The profits from timber sales are estimated to be reduced by over 10 million Euros annually due to the recreation-enhancing practices (Kosenius et al., 2013).

A few previous Finnish studies with different geographical scopes have considered public preferences for forest biodiversity, recreation and management.³ From a policy perspective, however, studies focusing on individual sites or small specific areas have limited relevance for decision making on a wider scale. For that purpose, aggregate benefits across all affected areas can be required (Heal et al., 2005; Hill and Courtney, 2006; Bestard and Font, 2010). The present study differs from the previous ones in terms of target population and focus. Focusing on the SOCFs in all of Finland, we consider public preferences for the recreation-enhancing management practices and the related benefits from these lands nationwide, and elaborate on the differences of these valuations across Finland.

This choice experiment focuses on three recreation-enhancing management practices that are commonly applied in the SOCFs: the width of *scenic buffer zones* along lakes and rivers, *habitats for game birds* (more specifically, managed courting grounds for the capercaillie, *Tetrao urogallus*), and the *quality of scenery* as reflected by the frequency of clear-cut areas along hiking trails. This study considers three types of policy questions. First, do the aggregate benefits from the recreation-oriented management regime as a whole exceed the associated opportunity costs? Second, what is the relative importance of the considered management attributes in the light of their marginal valuations? Third, what levels of the management attributes would maximize the benefits to the public? To answer these questions, we utilize conditional logit (CL) and random parameters logit (RPL) models (Train, 1998). The RPL (or mixed logit) is particularly fit for dealing with preference heterogeneity at the individual level across all attributes (Hynes et al., 2008).

Apart from the policy relevant empirical questions considered, the present study provides an application of a relatively new modeling approach. We estimate the marginal willingness-to-pay (WTP) effects for the management attributes using the RPL model specified directly in the WTP space rather than preference space. The WTP-space specification provides a number of advantages in panel mixed logit models (Train and Weeks, 2005; Sonnier et al., 2007; Scarpa et al., 2008; Thiene and Scarpa, 2009; Scarpa and Willis, 2010; Campbell et al., 2014).⁴ In particular, the fact that the coefficients directly provide marginal WTP estimates is very convenient for the present study, as we examine the importance of the management attributes and calculate the welfare effects of the current management regime for a cost-benefit analysis. To provide even more detailed management information, the most preferred levels of the management attributes are also considered in terms of physical measures. For this end, the CL model with quadratic sub-utility functions is estimated in the preference space.

The rest of the paper is organized as follows. Section 2 is devoted to the data and econometric models. Section 3 presents our empirical findings based on random parameters logit (RPL) models estimated in the WTP space to consider the benefits of management regimes, as well as a conditional logit (CL) model estimated in the preference space to consider the preferred levels of the management attributes in physical terms. Section 4 discusses the main findings, and Section 5 concludes the paper.

Material and methods

Survey design

The questionnaire was developed in spring 2012 in cooperation with Metsähallitus. The pilot survey was conducted as an on-site visitor survey in the Ylläs-Levi region in western Lapland. Around

³ These include visitor preferences for recreational and ecological management options for a national park (Juutinen et al., 2011), local recreationists' preferences for recreation and biodiversity in a set of municipal recreation sites (Horne et al., 2005), and citizens' as well as forest owners' preferences for safeguarding forest biodiversity (Horne et al., 2009). Tyrväinen et al. (2014) examine tourists' preferences for the landscape, outdoor routes, biodiversity, and carbon sequestration in a tourism area. Huhtala (2004) provides a contingent valuation study of the recreational value of all national parks and hiking areas. Travel cost models have been applied to recreation sites and hiking areas (Ovaskainen et al., 2001, 2012).

⁴ The WTP space specification accounts for scale variation, avoiding the potentially confounding effect of the scale factor and allowing the comparison of different models. Scale heterogeneity and the role of the scale parameter have attracted a lively interest recently (Swait and Louviere, 1993; Louviere and Eagle, 2006; Fiebig et al., 2010; Thiene et al., 2012). In addition, standard errors for the marginal WTPs are obtained without the distributional problems related to the ratios of two random coefficients, and the specification allows a control over the properties of WTP distributions.

Table 1

Attributes and their levels with variable names used in the analysis.

Attribute	Level	Variable name ^a
Width of buffer zones along lakes and rivers	1. 5 meters 2. 20 meters (current level) 3. 40 meters	Buffer5 Buffer40
Number of managed courting grounds for capercaillie	1. 1000 2. 2000 (current level) 3. 3000	CourtingGround1 CourtingGround3
Clear-cut areas visible along hiking trails (on % of trails)	1. Quite frequently (20%) 2. Quite rarely (10%, current level) 3. Not at all (0%)	Clearcut20 Clearcut0
Tax increase per household	0, 10, 30, 60, 100, 150 (€/year)	Tax

^a In the RPL model we used discrete attribute levels (effects coding) except that Tax was treated as a continuous variable. In the CL model also the non-cost attributes were treated as continuous with variable names Buffer, CourtingGround, and Clearcuts.

200 visitors filled in the pilot questionnaire. The final questionnaire was developed based on the findings, and consisted of four parts. The introductory part explored the respondent's general knowledge about and free time activities in SOCFs, and attitudes to the multiple use of SOCFs. The second part introduced the recreation-oriented management alternatives, and guided the respondents to consider the recreational experiences and the related forest management practices when answering the choice tasks in the following part. The third part contained the choice experiment, the related debriefing questions, and a set of attitudinal questions on the management of SOCFs. The final part included questions regarding respondents' socio-economic status.

The decisions regarding the attributes and their levels in the choice experiment (Table 1), as well as the implementation of the survey, were based on discussions with the management personnel and experts of Metsähallitus in the project steering group, existing statistics (Metsähallitus, unpublished), and experience from previous valuation studies (e.g., [Tyrväinen et al., 2010, 2014](#); [Juutinen et al., 2011](#)). The management attributes were chosen so as to be relevant for both policy makers and the public ([Bennett and Adamowicz, 2001](#)). These were the width of scenic buffer zones along lakes and rivers, habitats for game birds in terms of the number of managed courting grounds for the capercaillie, and the quality of scenery along the hiking trails. According to the experts of Metsähallitus, these three attributes represent the most typical management activities currently applied in SOCFs to enhance recreational services. Hence, these are also the activities that mainly cause the opportunity costs of recreation-oriented management. The courting grounds refer to habitats with mature pine forest surrounded by young (mixed) forest that provide favorable mating sites for the capercaillie. The courting grounds and buffer zones along lakes and rivers preserve wooded scenery in the landscape. The courting grounds also increase the opportunity to see wildlife in the forest during outdoor activities ([Boxall and Macnab, 2000](#)). As clear-cutting with soil preparation is the most commonly perceived negative landscape effect of forestry ([Ribe, 2005, 2009](#); [Silvennoinen et al., 2002](#); [Tyrväinen et al., 2014](#)), the quality of scenery along hiking trails was operationalized as the frequency of clear-cut regeneration areas visible along the trails. Before the pilot survey, the functioning of the questionnaire was tested by several individuals, thus allowing the pilot survey to concentrate on the attributes. The results of the pilot survey indicated that people regard the chosen management attributes as important factors for their recreational experience in SOCF, but also revealed some deficiencies in the descriptions of the attributes that were improved in the final questionnaire.

The status quo levels, based on the statistics of Metsähallitus (unpublished), refer to the nationwide average. The regional averages and individual respondents' experiences may differ from this. For each attribute, we specified the increase and the decrease in forest management activities, based on expert views on the potential future levels of recreation-oriented management in SOCFs.

To ensure that respondents have enough information for their choices and to mitigate hypothetical bias, the recreation-oriented management practice was described to respondents before the choice tasks. The respondents were first informed that Metsähallitus applies special management practices to enhance the provision of recreational services by SOCFs. The differences between the

Choice task 1.			
Attribute	Current situation	Alternative 1	Alternative 2
Width of buffer zones along lakes and rivers	20 m	40 m	20 m
Number of managed courting grounds for capercaillie	2000	1000	3000
Clear-cut areas visible along hiking trails	quite rarely	not at all	quite frequently
Tax increase per household	0 €/year	60 €/year	30 €/year
	 ☐	 ☐	 ☐

In my opinion, the best alternative is (please tick one circle).

Fig. 2. An example of the choice task.

recreation-oriented management and standard commercial management were illustrated by listing management activities used in the two regimes. It was also explained that recreation-oriented management practices reduce the profits of timber production that Metsähallitus remits to the government, and therefore, there is a pressure to increase taxes. Then the management attributes used in the choice experiment were described verbally and graphically, and the respondents were asked to consider what is the best management level for each attribute from the viewpoint of their own recreational experience and to choose the best level from among the given levels. Respondents were informed that the levels describe current management levels and potential future management levels for next 10 years. The same levels were used in the choice experiment.

The experimental design was created using a Bayesian efficient design with prior information from the pilot survey and optimized for D-efficiency for the RPL panel model (Ngene, 2010). The resulting 36 choice tasks were blocked into six groups. Thus, each respondent faced six choice tasks in which s/he selected between three alternatives presenting forest management options. Each choice task included two future scenarios for next 10 years and a status quo alternative, in which the levels of attributes refer to the current situation. Respondents were guided to choose the best alternative from their own point of view taking into account the fact that a tax increase reduces their disposable income. The tax increase was described as an annual increase in household taxes during next 10 years. Respondents were asked to consider the choice tasks as separate situations and answer each choice task. Fig. 2 presents an illustration of the choice task.

Data

The data were collected by a combination of mail and web surveys (a mixed mode) during summer 2012. In May 2012, we sent an invitation to participate in the web-based survey to 4200 randomly selected inhabitants of ages 18 – 75 years living in selected municipalities in three regions: Lapland, Kainuu, and Tavastia Proper. The regions are equivalent to the Nomenclature of Units for Territorial Statistics (NUTS, level 3) of the European Union. Fourteen hundred invitations were sent to each region. Three weeks and one reminder later we sent the second reminder accompanied with a printed questionnaire to non-respondents. The web opportunity was still available. Two weeks after the mail survey, we sent a reminder to non-respondents. In the end of July, we had received 1241 answers, making the response rate 29.5%. The majority (59.0%) of respondents answered the paper survey version. The region-specific response rate was highest in Kainuu (32.1%) followed by Lapland (28.9%) and Tavastia Proper (27.6%).

By addressing surveys to three regions, we aimed at covering citizens living at various socioeconomic situations and in areas with different opportunities to visit SOCFs. This would make the nationwide aggregation of the benefits more accurate. Table 2 presents the comparison of the three data sets and the corresponding populations in terms of socio-demographic characteristics. In most respects, the

Table 2

Comparison of samples and populations.

	Lapland		Kainuu		Tavastia Proper	
	Sample	Population	Sample	Population	Sample	Population
<i>N</i>	405	183,488	449	82,073	387	4,935,160
Proportion of males	48.0	50.2	52.0	50.0	48.0	49.3
Age (average)	51.6	51.3	53.3	52.7	52.7	49.6
Annual disposable household income (average, 1000€)	32.168	32.237	31.050	31.425	34.340	34.790
Annual household income before taxes (average, 1000€)	48.300	48.404	46.413	46.973	52.750	53.441
Proportion of households with dependent children	43.0	37.7	43.3	36.0	46.9	38.9
Households size (average)	2.39	2.08	2.43	2.05	2.46	2.09
High education (% of university degree)	8.6*	4.9	7.8*	4.4	10.9	8.7
Proportion of agricultural entrepreneurs	2.9	1.5	2.4	1.9	4.2**	1.6

Notes: The data on population are from Statistics Finland (2012). The sample of Tavastia Proper is compared with the population of Finland excluding populations from Lapland and Kainuu.

* 10% Significance level.

** 5% Significance level.

samples were not statistically significantly different from the population for which the results will be generalized.

Regarding the differing opportunities for forest visits, Lapland represents a region with plenty of SOCFs. Nature-based tourism has an important role in the region, and local people use the SOCFs actively. Kainuu as well represents a region with lots of SOCF where the large majority of the recreationists are local residents who participate very actively in close-to-home recreation. Tavastia Proper represents a southern region with almost no SOCF at all. Most people in Finland live in the latter type of region.

Models

The choice data were analyzed using conditional logit (CL) and random parameters logit (RPL) models. The former was used to elaborate on the respondents' preferred attribute levels with conventional specification of utility in the preference space and the latter was used to derive marginal willingness to pay estimates with specification of utility in the WTP space. CL and RPL models can be derived from a random utility model (McFadden, 1974), which assumes that the utility function of each respondent is the sum of a deterministic term (a function of factors that influence the respondent's utility), and a stochastic random term (unobservable to the researcher).

Consider first the RPL model, which captures taste heterogeneity across respondents, in the preference space. Denoting the cost attribute by p and the vector of non-cost attributes by x , the unobservable overall utility U of alternative i for respondent n is represented by:

$$U_{ni} = -(\alpha_n/k_n)p_{ni} + (\beta_n/k_n)'x_{ni} + \varepsilon_{ni}, \quad (1)$$

where α_n and β_n are the coefficients for the cost attribute and for the vector of non-cost attributes, respectively, k_n is the scale parameter, and ε_{ni} is an i.i.d. Gumbel distributed error term that has the same variance for all respondents. Defining $\lambda_n = \alpha_n/k_n$ and $c_n = \beta_n/k_n$ the utility can be rewritten as:

$$U_{ni} = -\lambda_n p_{ni} + c_n'x_{ni} + \varepsilon_{ni}. \quad (2)$$

Given Eq. (2), the implied WTP for a non-cost attribute is the ratio of the non-cost attribute's coefficient to the cost coefficient: $\omega_n = c_n/\lambda_n = \beta_n/\alpha_n$. Accordingly, the utility in WTP space can be presented as (Train and Weeks, 2005):

$$U_{ni} = -\lambda_n p_{ni} + (\lambda_n \omega_n)' x_{ni} + \varepsilon_{ni}, \quad (3)$$

in which the variation in WTP is independent of scale and separated from the variation in the cost coefficient, that incorporates scale.

The subscript n on λ and ω indicates that each coefficient varies across respondents in the population. Denote their joint density function by $f(\theta_n|\Omega)$, where θ_n represents the vector comprised of the random parameters and Ω represents the parameters of this distribution (e.g., mean and standard deviation in the population). Given the assumption of i.i.d. Gumbel distributed error, the unconditional choice probability is the integral of the logit formula over all possible values of λ_n and ω_n :

$$Pr(y_n|p_n, x_n, \Omega) = \int \prod_{t=1}^{T_n} \frac{\exp(-\lambda_n p_{nit} + (\lambda_n \omega_n)' x_{nit})}{\sum_{j=1}^J \exp(-\lambda_n p_{njt} + (\lambda_n \omega_n)' x_{njt})} f(\theta_n|\Omega) d(\theta_n), \quad (4)$$

where y_n gives the sequence of choices over T_n choice occasions for respondent n (Campbell et al., 2014).

In the empirical part of this paper we specify λ as having a Log-Normal distribution given the theoretical expectations of disutility for price and following previous literature on WTP-space models (e.g., Scarpa et al., 2008; Thiene and Scarpa, 2009; Campbell et al., 2014).⁵ For the distributions of ω we assume that they are all normally distributed. The non-cost attributes are effects coded, and the alternative specific constant (ASC) reflects the utility associated with the status quo alternative (current situation) per se, i.e., the factors not described by the considered attributes. The advantage of effects coding in comparison to dummy coding is that the effects of all levels of qualitative attributes can be estimated and that all estimates are uncorrelated with the intercept, or ASC (Bech and Gyrd-Hansen, 2005). Importantly, for each non-cost attribute, we can estimate the effect of the reference level which is the current level in our case (Holmes and Adamowicz, 2003). Thus, by applying the WTP-space specification and effects coding we can assess the monetary value for the current management level along with values for changed management levels. More specifically, the WTP values for the attribute levels corresponding to the current management can be derived from the two estimated WTP values of the attribute in question as a negative sum of these two values. Formally, the marginal WTP value of current management for attribute h is

$$MWTP_h = -1(MWTP_{h1} + MWTP_{h2}), \quad (5)$$

where subscripts 1 and 2 indicate the attribute levels associated with the changes from the current level. Denoting the number of considered attributes by H , the total value of the current level is $\bar{V}_C = \sum_{h=1}^H MWTP_h$.

The welfare measure for management changes can be derived from the compensating variation (CV) formula developed by Hanemann (1982), $CV = \frac{1}{\lambda}(V_0 - V_1)$, where V_0 and V_1 are the utility expressions for the base and altered management regimes, respectively. The formula can be rewritten as $CV = \frac{V_1}{-\lambda} - \frac{V_0}{-\lambda}$ that corresponds to the form we used in the WTP-space specification. Denoting $\bar{V}_0 = \frac{V_0}{-\lambda}$ and $\bar{V}_1 = \frac{V_1}{-\lambda}$, the compensating variation is calculated using the results of the WTP-space specification as

$$CV = \bar{V}_1 - \bar{V}_0. \quad (6)$$

In what follows, we use the management regime without recreation-oriented activities (i.e., with the levels of management attributes reduced from the current level; see Table 1) as the base case and the current management regime as the altered case. Hence, the calculated welfare effect measures

⁵ We specified $\lambda = -\exp(\mu + \sigma v)$, where v is a standard Normal deviate and μ and σ are the parameters to be estimated. BIOGEME version 2.0 (Bierlaire, 2003) was used in the estimation.

Table 3
Protest-related motives for serial nonparticipation and number of protesters by region and motive.

Motive	Lapland	Kainuu	Tavastia Proper
Choice situations were too difficult	8	5	6
Choice situations were not credible	3	2	2
I did not have enough knowledge on the issue	30	26	26
I do not accept a tax increase	27	36	36
In my opinion citizens should not be made to pay for recreation-enhancing forest management	16	25	25
Other reason	5	3	3
Total number of protesters (excluded from analysis)	89	97	94

benefits achieved by the current recreation-oriented management in a manner that is comparable with the opportunity costs of the current management.

The CL model can be derived from the RPL model by assuming that the coefficients for the attributes do not vary across respondents (Train, 2009). Even though this is a strong assumption, we use the CL model for an illustrative purpose to assess which levels of the management attributes are preferred by respondents, i.e., maximize the public benefits from recreation-oriented forest management in SOFCs. More specifically, we use the preference-space specification and define quasi-linear sub-utility functions for each non-cost attribute by using quadratic functions such that the sub-utility function for attribute h is $V_h = \beta_h x_h + \gamma_h x_h^2$. Thus in using the CL model we treat the non-cost attributes as continuous variables in contrast to the RPL model. The cost attribute is modeled by the first-order term only. After estimating the CL model, the optimal level (x_h^*) for the attribute h can be obtained by differentiating the sub-utility function and solving the first-order condition:

$$\frac{dV_h}{dx_h} = \beta_h + 2\gamma_h x_h = 0 \Rightarrow x_h^* = \frac{-\beta_h}{2\gamma_h}.$$

Results

Nonparticipation

The data show some degree of a frequently found response behavior, known as serial nonparticipation, whereby a number of respondents always choose the status quo (SQ) alternative (von Haefen et al., 2005). The approaches used to deal with the issue range from excluding these respondents to assigning them a priori to a specific class to using models that endogenously allow for different preference structures (see Thiene et al., 2012, and references therein). While serial nonparticipation is an interesting research issue, its detailed modeling is beyond the scope of the present paper. We therefore chose to use the information from a debriefing question concerning the motives for this behavior to identify groups with different motives (Table 3). Respondents that chose the SQ alternative (the current management regime) in each choice task and reported their motive to be a protest against the scenario or the payment vehicle, comprising 21.6 – 24.3% of all respondents, were removed from the data before further analysis. This is because such protest answers cannot be taken to reflect the respondent’s true valuation of the management alternatives per se. The respondents that always selected the SQ, but answered that this was truly the best alternative or that they cannot afford to pay more taxes, were not removed.⁶

⁶ A logit model analysis revealed that, in all regions, the probability of a protest answer increases with high age and low education, and if the respondent has not visited state-owned commercial forests during the last 12 months. In addition, several socio-economic variables, such as the respondent’s gender and agricultural entrepreneurship, have region-specific impacts on the probability of a protest answer.

Table 4

Results of random parameters logit (RPL) model in WTP space.

Attribute levels ^a		Lapland		Kainuu		Tavastia Proper	
		Coeff.	S.E.	Coeff.	S.E.	Coeff.	S.E.
Buffer5	Mean	−76.0***	11.5	−74.8***	9.5	−75.0***	7.7
	S.D.	48.6***	9.6	31.9***	9.7	38.0***	7.9
Buffer40	Mean	34.4***	9.0	55.7***	7.3	42.9***	6.8
	S.D.	40.6***	6.6	27.9***	5.6	30.2***	4.9
CourtingGround1	Mean	−39.5***	5.1	−38.8***	6.3	−29.1***	5.2
	S.D.	32.6***	8.1	31.9***	7.3	9.4	6.8
CourtingGround3	Mean	19.8***	6.6	21.2***	6.0	15.7***	5.1
	S.D.	20.1***	3.4	28.4***	6.5	5.4	12.6
Clearcut20	Mean	−40.4***	8.0	−32.4***	5.7	−38.6***	6.1
	S.D.	46.4***	8.5	17.9**	8.1	28.5***	4.6
Clearcut0	Mean	13.6**	6.5	13.5**	4.7	15.4***	4.9
	S.D.	47.1***	2.1	14.5***	6.1	27.3***	5.4
ASC (for current management)	Mean	38.0*	20.3	22.9	15.5	8.2	13.0
	S.D.	156.0***	17.8	199.0***	22.1	118.1***	13.6
Tax	Mean	−3.87***	0.125	−3.76***	0.123	−3.71***	0.111
	S.D.	0.534**	0.240	0.486***	0.205	0.604***	0.179
N:o of sets		1715		1920		1644	
Log Likelihood		−1143.55		−1318.54		−1152.12	
Pseudo R ²		0.39		0.38		0.36	

^a Effects coded.

* Statistically significant at 10% levels.

** Statistically significant at 5% levels.

*** Statistically significant at 1% level.

Marginal WTP values and importance of management attributes

In the RPL models for all three data sets, all coefficients except the ASCs⁷ are statistically significant at the 10% level (Table 4). The parameters have the expected signs. Compared to the current levels, decreases (increases) in the recreation-oriented forest management attributes reduce (increase) the probability of choosing an alternative. The opposite holds for the price attribute. The majority of the estimated coefficients for standard deviations are statistically significant, implying heterogeneity across respondents for forest management practices. The pseudo R^2 values (0.36 – 0.39) indicate a good model fit. Moreover, comparison to the lower pseudo R^2 statistics (0.23 – 0.25) and log-likelihood values (−1396.36 to −1614.19) of the conditional logit (CL) models in WTP space that served as a starting point for the analysis (not reported) indicates the superiority of the reported RPL models in terms of model fit.

Table 4 presents the marginal WTPs for increases and decreases in recreation-oriented management practices. The greater the absolute value of marginal WTP, the more important is the attribute

⁷ The status quo parameter (ASC) can convey information on a variety of different effects (see Thiene et al., 2012). As the ASCs are positive, they indicate some preference for the current situation. However, the ASCs are not statistically significant except in Lapland at the 10% level suggesting that the included attributes actually capture a remarkable part of features regarded important by the respondents. The fact that those respondents who always choose the SQ as a protest against the scenario or the payment vehicle were removed from the data may contribute to reduce the likelihood of a positive ASC. Importantly, the ASCs have quite large coefficients for the standard deviations. This may reflect the fact that we could not fully control for the nonparticipation.

for the respondents. Considering improvements and positive WTP values first, the citizens typically put the highest value on an increase in the width of buffer zones along lakes and rivers, followed by an increase in the number of managed courting grounds for capercaillie (i.e., habitats for game birds) and a decrease in the frequency of clear-cut areas visible along hiking trails. Regarding reductions in recreation-oriented management, as indicated by the negative WTP values, citizens would require quite high compensation levels to retain their welfare unchanged. The highest compensation value is estimated for the decrease in the width of buffer zones along lakes and rivers. Generally, the results indicate that it is very important to the citizens that the management activities are not reduced from the current level as the absolute values of reductions are much larger than the values of increased management activities. In addition, the width of buffer zones along lakes and rivers is the most important of the three attributes.

Interestingly, public preferences differ by the region (Table 4).⁸ Compared to citizens in northern Finland (Lapland) and southern Finland (Tavastia Proper), the citizens in eastern Finland (Kainuu) put a higher value on increase in the width of buffer zones along lakes and rivers. This means that the residents in Kainuu stress the importance of buffer zones wider than currently. The citizens in northern and eastern Finland put a higher value on an increase in the number of managed courting grounds for capercaillie than the citizens in southern Finland. In contrast, the citizens in southern Finland have the smallest compensation claims for decreases in the number of courting grounds. The citizens in eastern Finland have the lowest compensation claims for increases in the clear-cut areas visible along hiking trails.⁹

Aggregate benefits from the current management regime

Environmental valuation methods are typically used to assess the monetary benefits from a quality improvement of a good. In the present case, however, we focus on the benefits from the current management regime in order to conduct a cost-benefit comparison. Information on the costs of recreation-oriented management was available only for the current situation. According to Metsähallitus, the costs of the current recreation-oriented management are about 13 million annually (Kosenius et al., 2013). The cost estimate reflects opportunity costs, because it is calculated as the difference between the maximum net harvest revenues obtained without recreation-oriented management practices and with restrictions due to these practices by using data on stand characteristics and a forest growth simulation model named MELA (Redsven et al., 2012).

Regional differences in preferences for increases and decreases in management activities reflect the citizens' valuations of the current management regime in the regions (Fig. 3). Regarding citizens in Lapland, for example, the value of decreased (increased) buffer zone along lakes and rivers is €–76.0 (€34.4) as shown in Table 4, so the value of the current level for this attribute is €41.6 (Eq. (5)). Taking into account all the considered attributes the values of the current regime are €88.1, €55.6, and €68.7 per household per year for Lapland, Kainuu, and Tavastia Proper, respectively. The citizens in eastern Finland (Kainuu), for example, place the lowest value to the current management regime indicating

⁸ A simple *t*-test suggests that the differences are statistically significant, but see Poe et al. (2005) for more advanced tests and caveats related to simple computational methods including normality-based tests. The more advanced tests are computationally intensive and beyond the focus of this paper.

⁹ Several socio-economic characteristics, specified as interactions with attribute levels in the RPL model, were statistically significant in explaining citizen's preferences (results available upon request from the authors). Typically the effects of socio-economic characteristic were region-specific. For example, older respondents in Lapland more likely chose the alternative with less courting grounds and more frequent clear-cut areas visible along hiking trails. Male respondents in Tavastia Proper chose more likely the alternative with more courting grounds, and with wider buffer zones. In contrast, male respondents in the Lapland sample more likely chose the alternative with more frequent clear-cuts along hiking trails. People with university degree less likely chose an alternative with no clear-cuts along hiking trails in Lapland, and an increase in household income increased the probability to choose an alternative with more managed courting grounds in Lapland and Tavastia Proper. Along with the socio-economic characteristics we tested if the survey mode (mail, web) had any effect on the results. Regarding Lapland, for example, there was weak evidence (10% significance level) that respondents to the web survey placed a higher emphasis on the clear-cut areas along hiking trails than the mail respondents. Thus the characteristics of mail and web respondents differed, but after controlling the socio-economic characteristics, the survey mode did not seem to have any significant effect on the results.

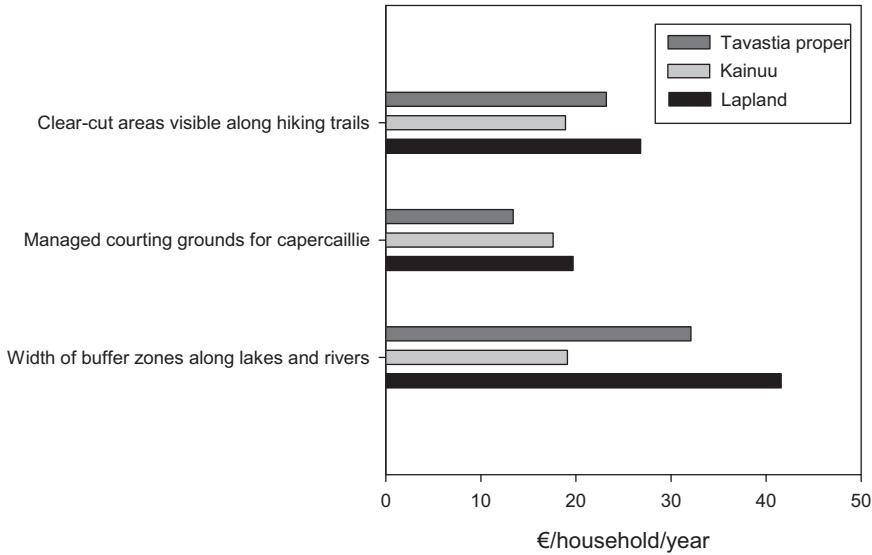


Fig. 3. Average willingness-to-pay values for the current levels of the forest management attributes. Lapland, Kainuu, and Tavastia Proper represent northern, eastern, and southern Finland, respectively.

that they typically would like to increase the recreation-oriented management activities more than the citizens in the other two regions did. In particular, this concerns the width of buffer zones along lakes and rivers.

Recall that the welfare effect of the current management regime (€/household/year) is calculated as a difference between the total value of the current regime and total value of the scenario in which the recreation-enhancing management practices are abandoned, i.e., the management attributes are at their lowest levels (Eq. (6)). To derive a conservative estimate for the aggregate welfare effect we assume that the compensating variation (CV) is zero for a share of citizens equal to the nonresponse rate of our questionnaire in each region. Multiplying the regional mean CV values by the number of households in the respective regions, with Tavastia Proper representing the rest of Finland except for Lapland and Kainuu, reveals that the aggregate welfare effect of the current recreation-oriented management practices is about €149 million per year. The estimated monetary benefits are much higher than the annual opportunity costs of about €13 million. Thus, the current management regime clearly passes the standard compensation test of cost-benefit analysis.

The most preferred levels of management attributes

Apart from the monetary benefits from the current management regime as a whole and the valuation of changes in specific attributes, an interesting question concerns the most preferred levels of the management attributes from the respondents' point of view. To illustrate this, we interpret the management attributes as continuous variables and, using the numerical values of attribute levels (see Table 1), estimate a conditional logit (CL) model with a quasi-linear utility function including both first-order and second-order terms for the management attributes. The estimated parameters are statistically significant except for the first-order term for the frequency of clear-cut areas along the hiking trails in northern and eastern Finland (Table 5).

Using the estimated parameters, we define the most preferred attribute levels by differentiating each attribute's sub-utility function and solving the first order condition. Table 6 shows the most preferred levels for forest management attributes in three regions. The results should be interpreted with some caution, as the coefficients of the first-order term for the frequency of clear-cut areas along the hiking trails for northern and eastern Finland were not statistically significant. In general, however,

Table 5

Results of conditional logit (CL) model with quasi-linear utility function.

Attributes ^a	Lapland		Kainuu		Tavastia Proper	
	Coefficient	S.E.	Coefficient	S.E.	Coefficient	S.E.
Buffer	0.133 ^{***}	0.022	0.112 ^{***}	0.023	0.151 ^{***}	0.023
Buffer ²	−0.002 ^{***}	<0.000	−0.001 ^{***}	<0.000	−0.002 ^{***}	<0.000
CourtingGround	0.157 ^{***}	0.052	0.178 ^{***}	0.047	0.120 ^{**}	0.050
CourtingGround ²	−0.003 ^{***}	0.001	−0.004 ^{***}	0.001	−0.002 [*]	0.001
Clearcut	0.029	0.023	0.026	0.022	0.048 [*]	0.023
Clearcut ²	−0.002 ^{**}	0.001	−0.002 ^{**}	0.001	−0.004 ^{***}	0.001
Tax	−0.011 ^{***}	0.001	−0.012 ^{***}	0.001	−0.013 ^{***}	0.001
ASC (for current management)	0.295	0.191	0.509 ^{***}	0.183	0.067	0.188
N:o of sets	1715		1920		1644	
Log Likelihood	−1407.71		−1614.19		−1396.36	
Pseudo R ²	0.25		0.24		0.23	

^a Treated as continuous variables. The number of managed courting grounds for capercaillie was divided by 100 to scale this coefficient to the same level as the other coefficients.

* Statistically significant at 10% levels.

** Statistically significant at 5% levels.

*** Statistically significant at 1% level.

Table 6

The most preferred levels of management attributes compared to the current management levels.

Attributes	Preferred attribute levels			Current level
	Lapland	Kainuu	Tavastia Proper	
Width of buffer zones along lakes and rivers (meters)	28.7	39.3	31.0	20.0
Number of managed courting grounds for capercaillie	2432.7	2480.1	2648.9	2000.0
Clear-cut areas visible along hiking trails (on % of trails)	6.0	5.6	6.6	10.0

the most preferred levels suggest that people would like to increase the management activities from the current level to some extent, particularly in the Kainuu region. Citizens in Kainuu prefer wider buffer zones along lakes and rivers and less frequent clear-cut areas along hiking trails than citizens in Lapland and Tavastia Proper. In contrast, the citizens in Tavastia Proper prefer the largest number of managed courting grounds for capercaillie.

Discussion

This study elicited the public preferences for recreation-oriented forest management in the state-owned commercially managed forests in Finland. Using the choice experiment method and data from three regions in Finland the study estimated the monetary benefits from the less intensive forest management regime currently applied to enhance forest recreational benefits. The considered forest management options were the width of buffer zones along lakes and rivers, number of managed courting grounds for capercaillie (*Tetrao urogallus*), and frequency of clear-cut areas visible along hiking trails. In addition, the most preferred levels for the considered management attributes were estimated.

Elaborating the marginal WTP values of the three attributes we found that scenic buffer zones along lakes and rivers was the most important attribute for the citizens. Wider buffer zones along lakes and rivers had a clear positive effect on the citizens' welfare. In contrast [Holmes and Boyle \(2003\)](#) found that riparian buffers had no significant effect. Notice, however, that rather than being riparian water protection zones required by the Water Act, the buffer zones in our case were based on scenic considerations. The scenic aspect of the buffer zones is directly observable to the public, and likely emphasized in a country with a lot of lakes like Finland.

Our results showed that less frequent clear-cut areas visible along hiking trails had a positive effect on the citizens' welfare. Similarly, [Tyrväinen et al. \(2014\)](#) showed that improvements in landscape

quality in terms of less frequent sights of clear-cut and site preparation areas along hiking trails had a clear positive effect. Our result is also in line with previous scenic preference studies suggesting that clear-cuts with artificial regeneration is commonly perceived as negative from the scenic point of view (e.g., Ribe, 2005, 2009; Silvennoinen et al., 2002) as well as with CE studies that have documented effects of clear-cutting (Hanley et al., 1998; Boxall and Macnab, 2000; Holmes and Boyle, 2003).

In addition, our results showed that providing habitats for game birds in terms of increased numbers of courting grounds for the capercaillie had a positive effect. The result is in line with Boxall and Macnab (2000) where the opportunity to see wildlife had a positive impact on trip values especially for wildlife viewers and evidence of moose populations (similar to Adamowicz et al., 1997, 1998) had a positive value especially for moose hunters. Notice that the habitats for capercaillie were not meant to represent biodiversity (species richness) considerations but an opportunity to see a capercaillie (especially the cock), an element of the forest landscape, and an option to hunt the capercaillie (in the future).

Along with marginal WTP values, we presented an estimate of aggregate benefits reflecting Finnish people's valuation of the current recreation-oriented management regime. Using a conservative estimate, we found that the value of benefits associated with the current recreation-oriented management practices is about €149 million per year. The current management regime clearly passes the standard compensation test of cost-benefit analysis, because estimated monetary benefits exceed the annual opportunity costs of about €13 million. This result is in line with earlier findings. For example, Huhtala (2004) estimated that recreational monetary benefits provided by Finnish national parks and state-owned hiking areas exceed the annual expenditures on maintenance.

There were regional differences in the public preferences. People in northern Finland seem to be quite pleased with the current situation in SOCFs as they placed the highest value to the current management regime. State-owned forests are the dominant forest type in Lapland. Therefore, it is not a surprise that northern residents put a high value on recreation services provided by SOCF. In contrast, the residents of eastern Finland placed the lowest value to the current management regime indicating that they typically would like to increase the recreation-oriented management activities more than people in the other two regions did. In particular, this concerns the width of buffer zones along lakes and rivers. This outcome likely reflects the fact that there are quite a lot of lakes and rivers in eastern Finland. The valuation of the residents of southern Finland is likely to reflect option values. Southern residents make quite actively nature based travels to northern Finland (Sievänen and Neuvonen, 2011).

From the management viewpoint, the results imply that people seem to be satisfied with the current management regime. The current degree of management is valued quite highly, thus supporting the recreation-oriented management in the SOCFs. Still, as showed by our illustration of the preferred levels of management practices, people would prefer an increase in the considered recreation-enhancing forest management activities in comparison to the current management regime. This result is contingent on the assumption of quasi-linearity for the utility function. For example, this modeling approach does not take into account a possible loss aversion related to the attributes that include both reductions and increases from the current level (Hess et al., 2008). Determining the socially optimal degree of recreation-oriented management would require detailed information about the marginal costs of management practices, which was not available.

The focus in this research was on recreation-oriented forest management practices, and our results suggest that the selected attributes captured comprehensively the features regarded important by the public. However, our briefing questions revealed that some respondents did not know very well these forest management activities. The capercaillie attribute in particular seemed to be difficult, and it is possible that some respondents have mixed the intended recreational and scenic aspects of this attribute with forest biodiversity issues (Jacobsen et al., 2008). Considering potential further analyses, the data also allows the investigation of inferred and stated attribute non-attendance, thus deepening the understanding on the importance of certain attributes (Scarpa et al., 2013). To shed more light on preference heterogeneity and to identify segments of citizens that may have conflicting interests regarding the management of SOCFs, the choices could be modeled with the latent class model (e.g., Tolvanen et al., 2013).

Conclusions

Forests used for commercial timber production also provide important recreational services to the public in many countries. These recreational benefits are affected by the adopted forest management practices that change the forest scenery and may hamper outdoor activities in the forests. The width of scenic buffer zones along lakes and rivers turned out to be a particularly important forest attribute for the Finnish citizens in this study. More generally, forest management practices that preserve the wooded scenery increase people's welfare. Our cost-benefit analysis showed that the benefits of recreation-oriented practices were clearly higher than the associated opportunity cost. The result supports the conclusion that forest managers should use recreation-enhancing practices in areas that are regarded important from the recreation viewpoint. This is particularly true in managing public forest land, but there is also a need to develop a payment mechanism for private forest owners to compensate the losses caused by recreation-oriented practices. We developed a modeling approach to assess citizens' preferred levels of management attributes. The results indicate that the width of buffer zones along lake and rivers should be increased from the current level especially in the lake-rich Kainuu region. Thus, it is important to take into account regional differences in targeting the recreation-oriented practices.

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